

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CG

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INDEX NUMBER

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

12 September 2018

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CG and index number on page 1, 5, 9 and 13 in the spaces provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	15
2	20
3	15
4 or 5	25
Total	75

This document consists of **18** printed pages.



Section A

Answer **all** the questions in this section.

- 1 Red blood cells are formed from cells called reticulocytes. Stem cells in the bone marrow produce reticulocytes which differentiate into red blood cells. Red blood cells have an average life span of 120 days, after which they are destroyed in the liver.

(a) Describe how stem cells in the bone marrow produce red blood cells.

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..... [2]

New red blood cells are constantly produced in the bone marrow as stem cells, unlike other somatic cells, can divide throughout the lifespan of the individual. Somatic cells undergo a limited number of cell divisions as the DNA daughter strand is shortened with every replication.

Fig. 1.1 shows the shortened daughter strand with its parent template after a replication.



Fig. 1.1

(b) (i) Explain why the daughter DNA strand is shortened after each replication.

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..... [2]

(ii) Explain why stem cells are able to divide throughout the lifespan of the individual.

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..... [2]

During differentiation, haemoglobin is produced. Fig. 1.2 shows the structure of a section of the haemoglobin gene and its corresponding messenger RNA (mRNA) in the nucleus of a reticulocyte during transcription.

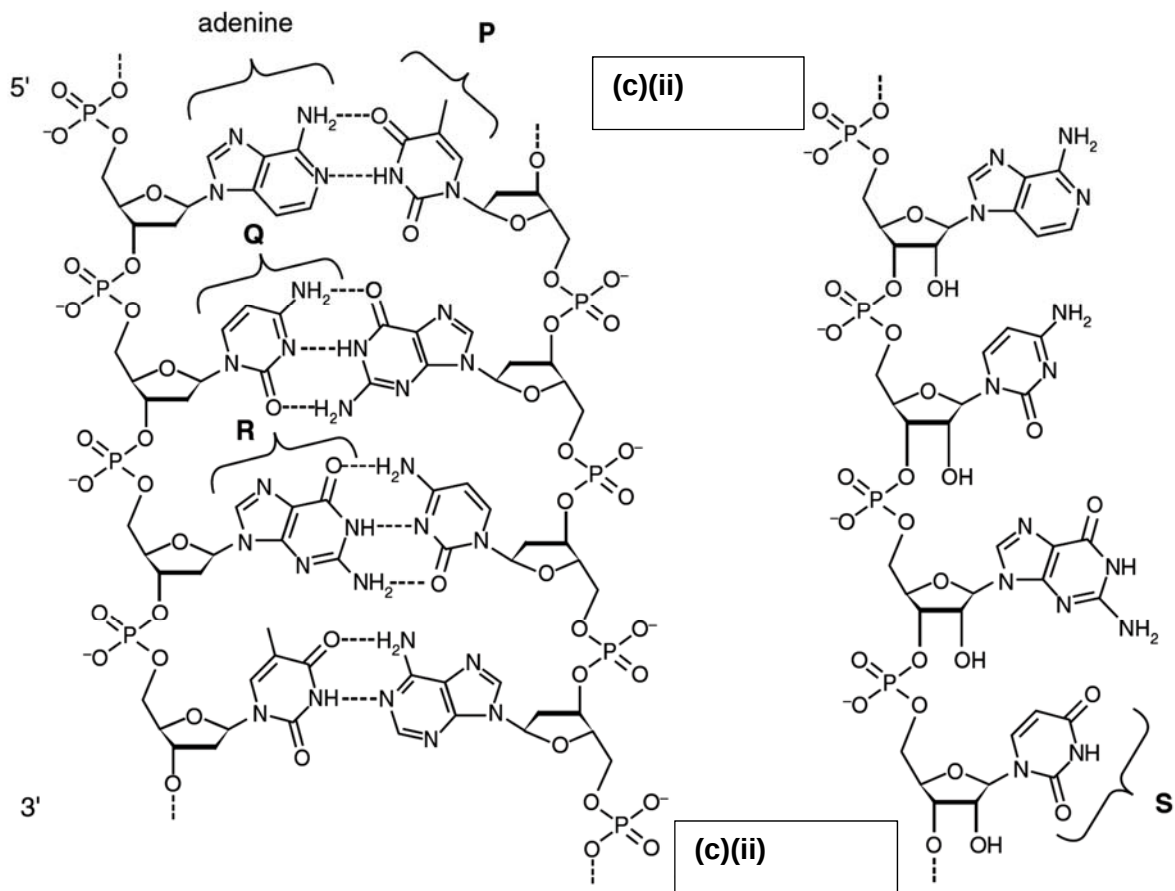


Fig. 1.2

(c) (i) Identify structures **P – S**.

P

Q

R

S

[2]

(ii) On Fig. 1.2, label the 5' and 3' end of the mRNA molecule.

[1]

- (iii) Briefly describe the fate of the mRNA molecule after transcription before its export out of the nucleus.

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..... [3]

- (d) Describe **two** differences between transcription and DNA replication.

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..... [2]

- (e) Suggest **one** additional change, besides production of haemoglobin that occurs to produce reticulocytes during their differentiation into red blood cells.

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..... [1]

[Total: 15]

- 2 Lactate dehydrogenase (LDH) is an enzyme found in many organisms. Within the same organism, it can be found in different forms, called isoenzymes. The isoenzymes are structurally different but they all catalyse the same reaction.

Fig. 2.1 shows a reaction catalysed by lactate dehydrogenase that occurs during anaerobic respiration in muscle tissue.

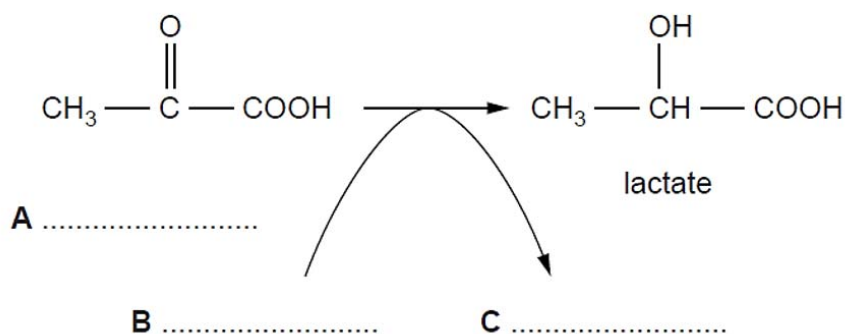


Fig. 2.1

(a) With reference to Fig. 2.1,

- (i) identify compounds A, B and C.

A

B

C [2]

- (ii) state the location in the cell where this reaction takes place.

..... [1]

- (iii) explain the significance of this reaction in mammalian muscle tissue.

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 [3]

Lactate dehydrogenase isoenzymes are globular proteins, each consisting of four polypeptides.

(b) Explain how the structure of an enzyme, such as lactate dehydrogenase is suited to its role.

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[4]

Lactate dehydrogenase isoenzymes are made up of two types of polypeptide: polypeptide M, which is coded for by the *LDH-A* gene and polypeptide H, which is coded for by the *LDH-B* gene.

Table 2.1 shows the composition of different human lactate dehydrogenase isoenzymes and examples of tissues and organs where each can be found.

Table 2.1

isoenzyme	polypeptide composition of enzyme	example of isoenzyme location
LDH-1	HHHH	heart red blood cells
LDH-2	HHHM	heart red blood cells
LDH-3	HHMM	brain lungs
LDH-4	HMMM	kidneys placenta
LDH-5	MMMM	liver skeletal muscles

(c) With reference to Table 2.1, suggest how red blood cells of the same individual can produce different isoenzymes.

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[2]

Besides lactate dehydrogenase, another extensively studied protein which is also involved in respiration is cytochrome c. Cytochrome c plays an important role in oxidative phosphorylation as an electron carrier of the electron transport chain.

Table 2.2 shows the amino acid sequence of a section of the cytochrome c polypeptide chain retrieved from 1 individual from each of the species below. The dashes shown in the figure indicates that the amino acid present at the position is identical to that of the human species.

Table 2.2

		1					6				10				14			17	18		20		
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His	Thr	Val	Glu	Lys
Pig		-	-	-	-	-	-	-	-	-	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	-
Chicken		-	-	Ile	-	-	-	-	-	-	-	Val	Gln	-	-	-	-	-	-	-	-	-	-
Dogfish		-	-	-	-	-	-	-	-	Val	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	Asn
Drosophila	<<<	-	-	-	-	-	-	-	-	Leu	-	Val	Gln	Arg	-	Ala	-	-	-	-	-	-	Ala
Wheat	<<<	-	Asn	Pro	Asp	Ala	-	Ala	-	-	-	Lys	Thr	-	-	Ala	-	-	-	-	-	Asp	Ala
Yeast	<<<	-	Ser	Ala	Lys	-	-	Ala	Thr	Leu	-	Lys	Thr	Arg	-	Glu	Leu	-	-	-	-	-	-

(d) With reference to Table 2.2,

- (i)** suggest what information indicates about the evolutionary relationships between humans and the other species;

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..... [3]

- (ii)** explain why the stated conclusion in **(d)(i)** needs to be treated with caution.

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..... [2]

- (e)** Suggest how the differences in the amino acid sequences shown in Table 2.2 may have arisen.

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..... [2]

- (f) Suggest why the cytochrome c protein was chosen to compare amino acid sequences across the different species.

----- [1]

[Total: 20]

- 3 A student cut thin sections of a root tip of *Allium cepa* (*A. cepa*) and stained them to show chromosomes. A photomicrograph of part of one of these sections is shown in Fig. 3.1.

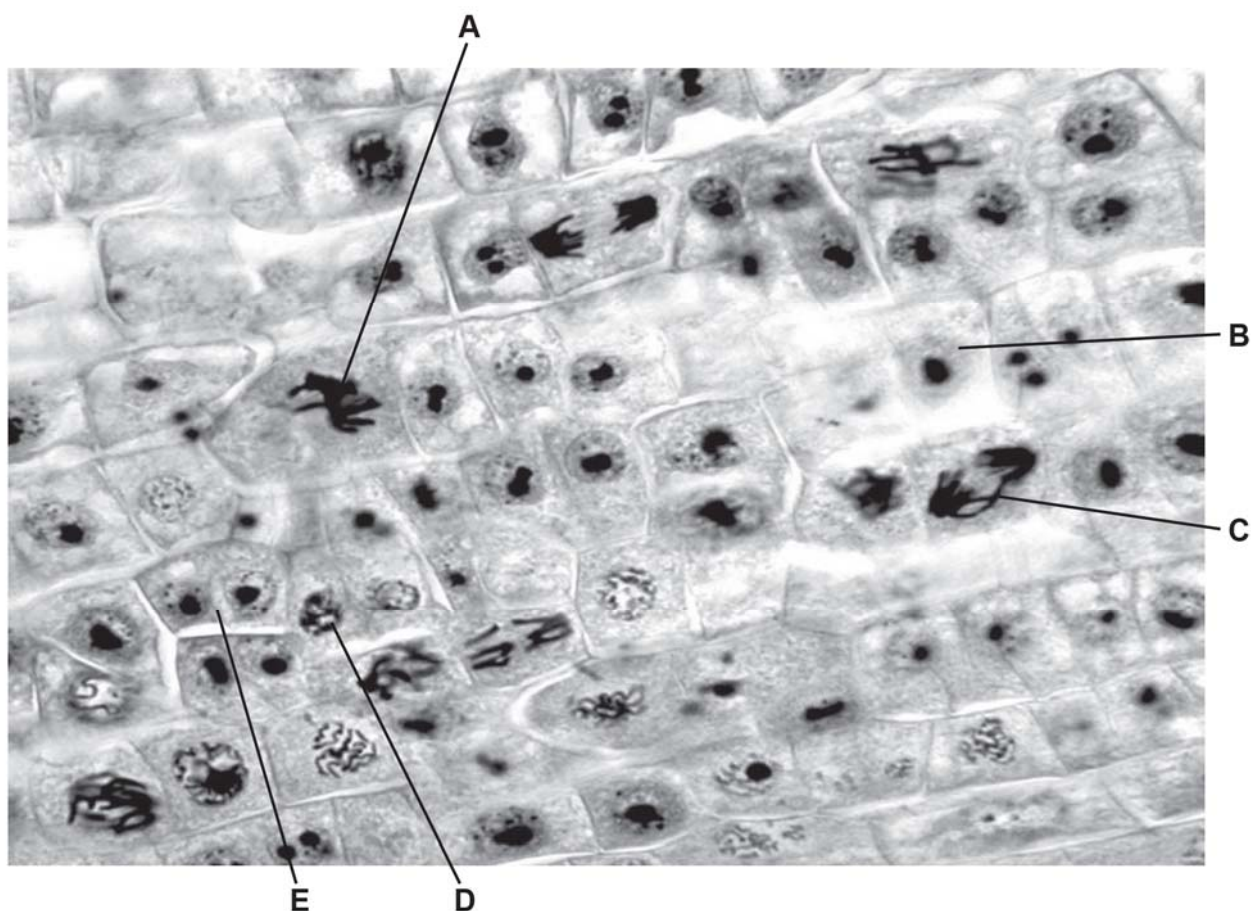


Fig. 3.1

- (a) Table 3.1 shows the behaviour of chromosomes and the changes that occur to the nuclear envelope during a mitotic cell cycle in the root tip of *A. cepa*.

Complete Table 3.1.

Table 3.1

name of stage	letter of cell	behaviour of chromosomes	nuclear envelope
	B	uncondensed, may be replicating	
prophase			intact, eventually breaks down
metaphase			Not present
anaphase			Not present
telophase		uncoil/ to become thin chromatin threads	

[5]

- (b) Explain why the growth of roots, such as those of *A. cepa*, involves mitosis and not meiosis.

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..... [3]

- (c) Some bacteria infect the plant to result in tumour formation at the infected tissues as shown in Fig. 3.2.

Research on the tumour formation processes revealed that there was an unusual excessive production of auxin, an important plant growth hormone. Auxin has been found to regulate gene expression resulting in cell division, expansion, differentiation and specialisation.

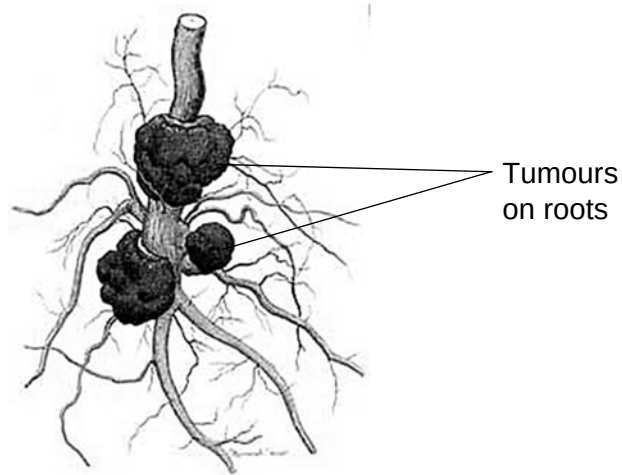


Fig. 3.2

Suggest how the bacteria infection resulted in tumour formation.

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..... [3]

- (d) A protein, mitosis promoting factor (MPF), has been identified in cells and has a role in initiating prophase.

MPF is made up of cyclin B1 coupled with cyclin-dependent kinase 1 (CDK1) and normally begins to break down and stops functioning during anaphase.

An overexpression of cyclin B1 has been reported in various human tumours, and is related to aneuploidy and high growth rate for cancerous growths in the breast.

- (i) Suggest reasons for the overexpression of cyclin B1.

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..... [3]

- (ii) State what does *aneuploidy* mean.

..... [1]

[Total: 15]

Question 4 and 5 starts on Page 13

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a) ATP is known as the universal currency of energy in living organisms. It can be utilised and regenerated in various chemical and physical reactions that occur in living systems.

With reference to named examples, outline the roles of ATP in living organisms. [13]

- (b) Plasma membranes exist on the surface of cells and in the endomembrane systems within the cells. They play critical roles in the regular functioning of a cell.

Discuss the functions of various components in plasma membrane and explain why there is a different composition of these components in membranes of different cells and organelles. [12]

[Total: 25]

- 5 The immune system responds to foreign antigens by producing specific immune responses to remove the antigen.

- (a) Outline the processes involved in the generation of antibody diversity which allowed the humoral immunity to be effective in both the primary and secondary response. [12]

- (b) The human immunodeficiency virus (HIV) is a retrovirus that causes AIDS. The infected person has a weakened immune system and is susceptible to opportunistic infections which can lead to their death.

Describe the HIV reproductive cycle and explain how these opportunistic infections may arise. [13]

[Total: 25]

